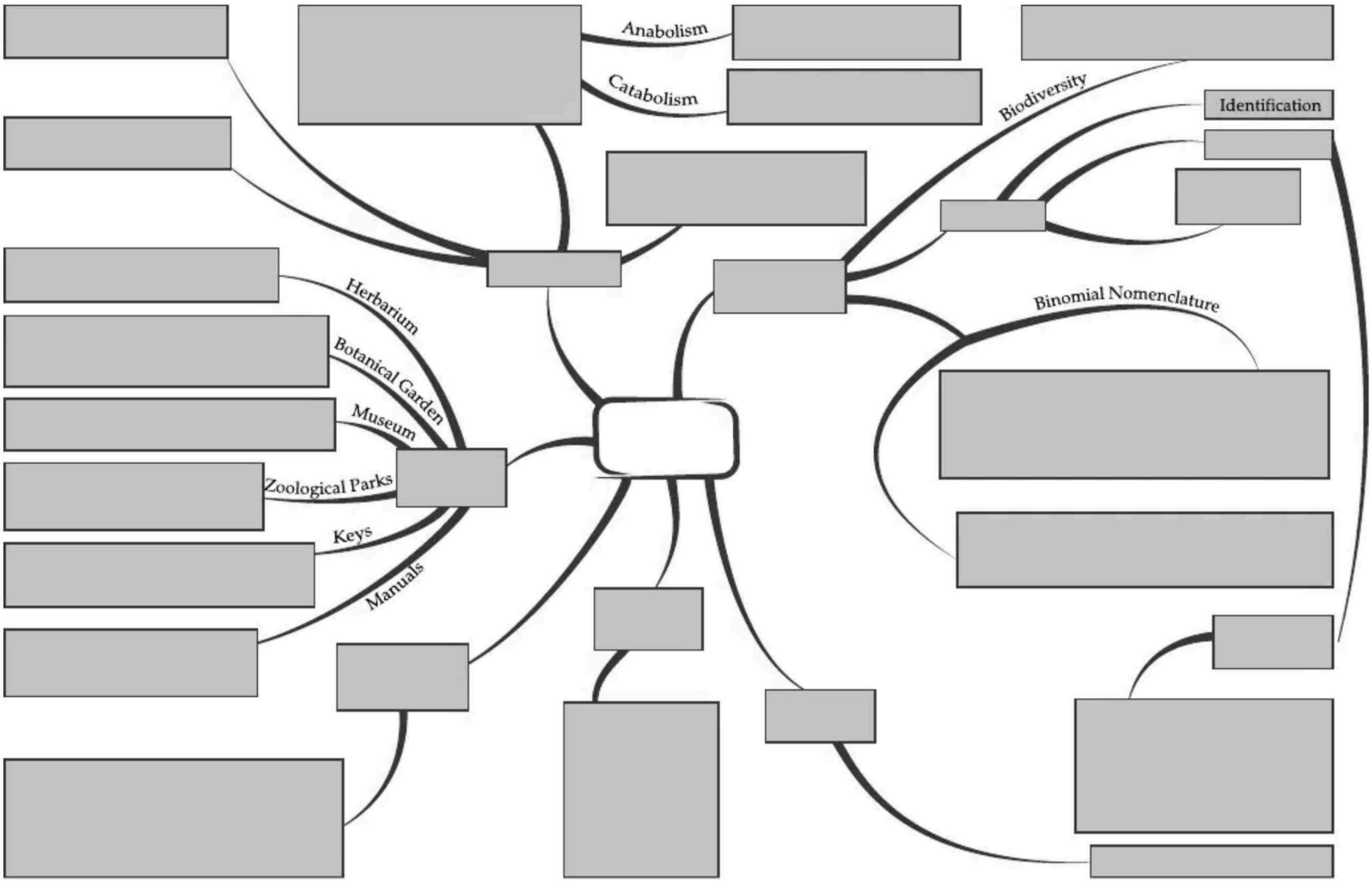
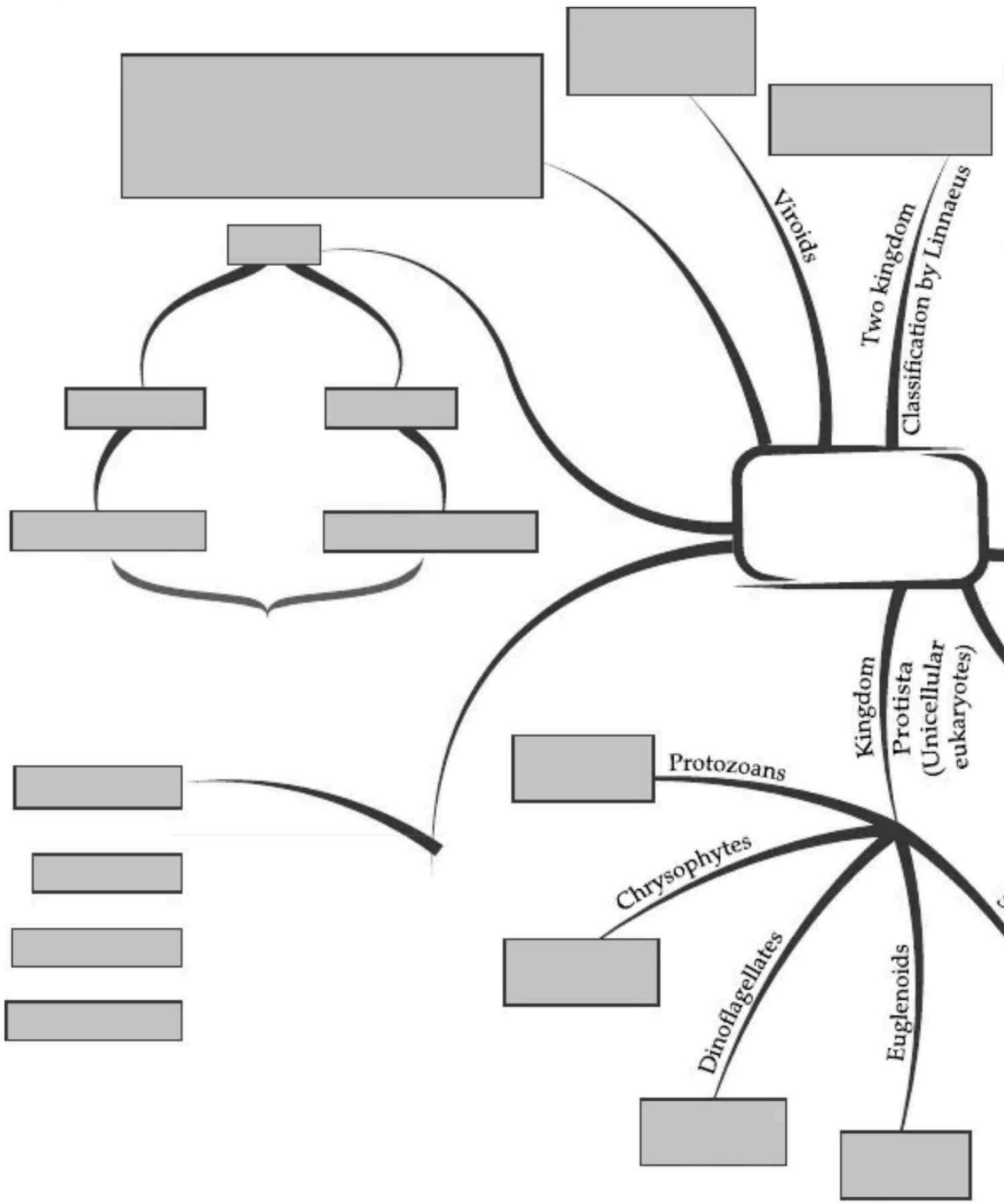
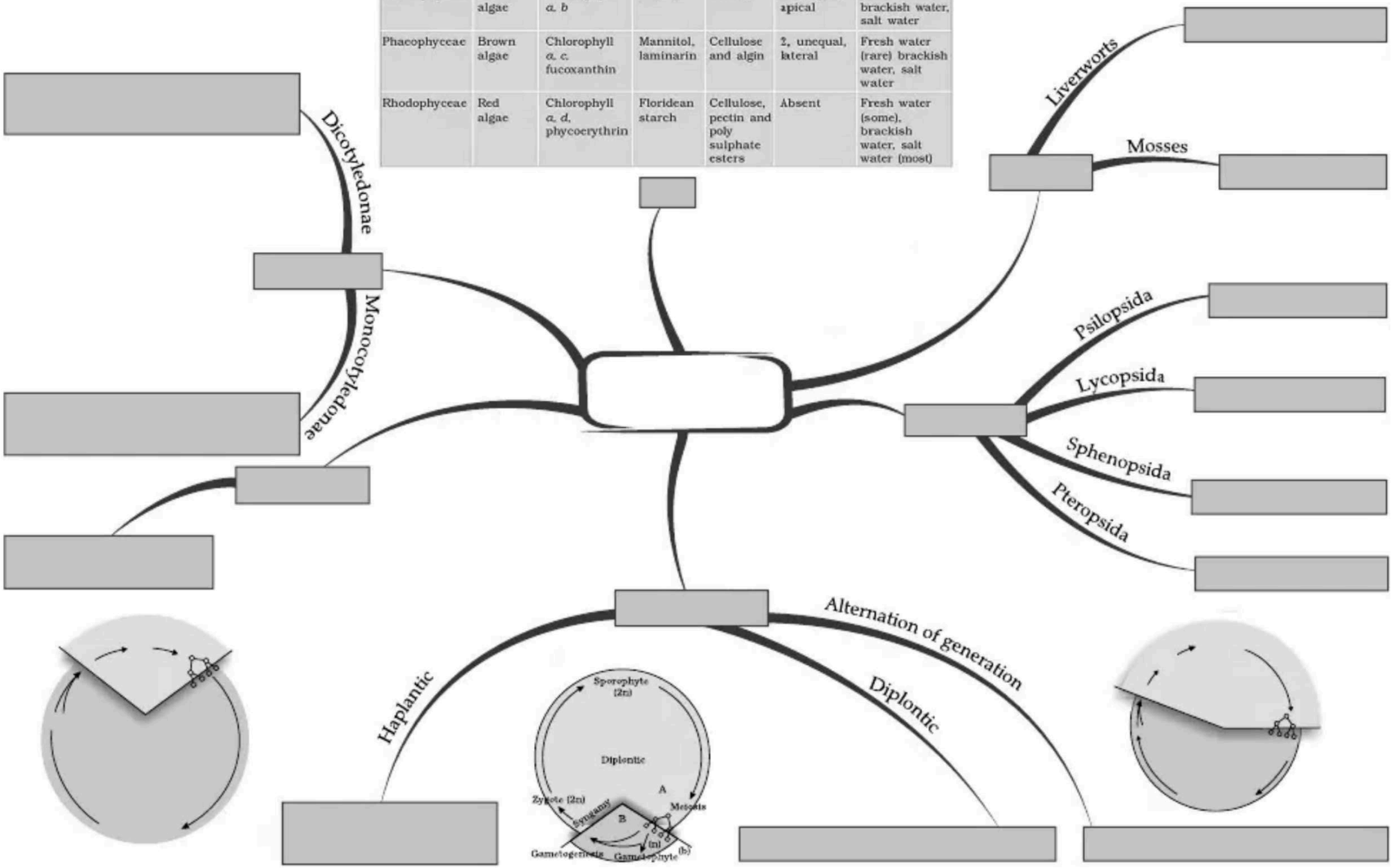

Anabolism
Catabolism
Biodiversity
Identification
Herbarium
Botanical Garden
Museum
Zoological Parks
Keys
Manuals
Binomial Nomenclature

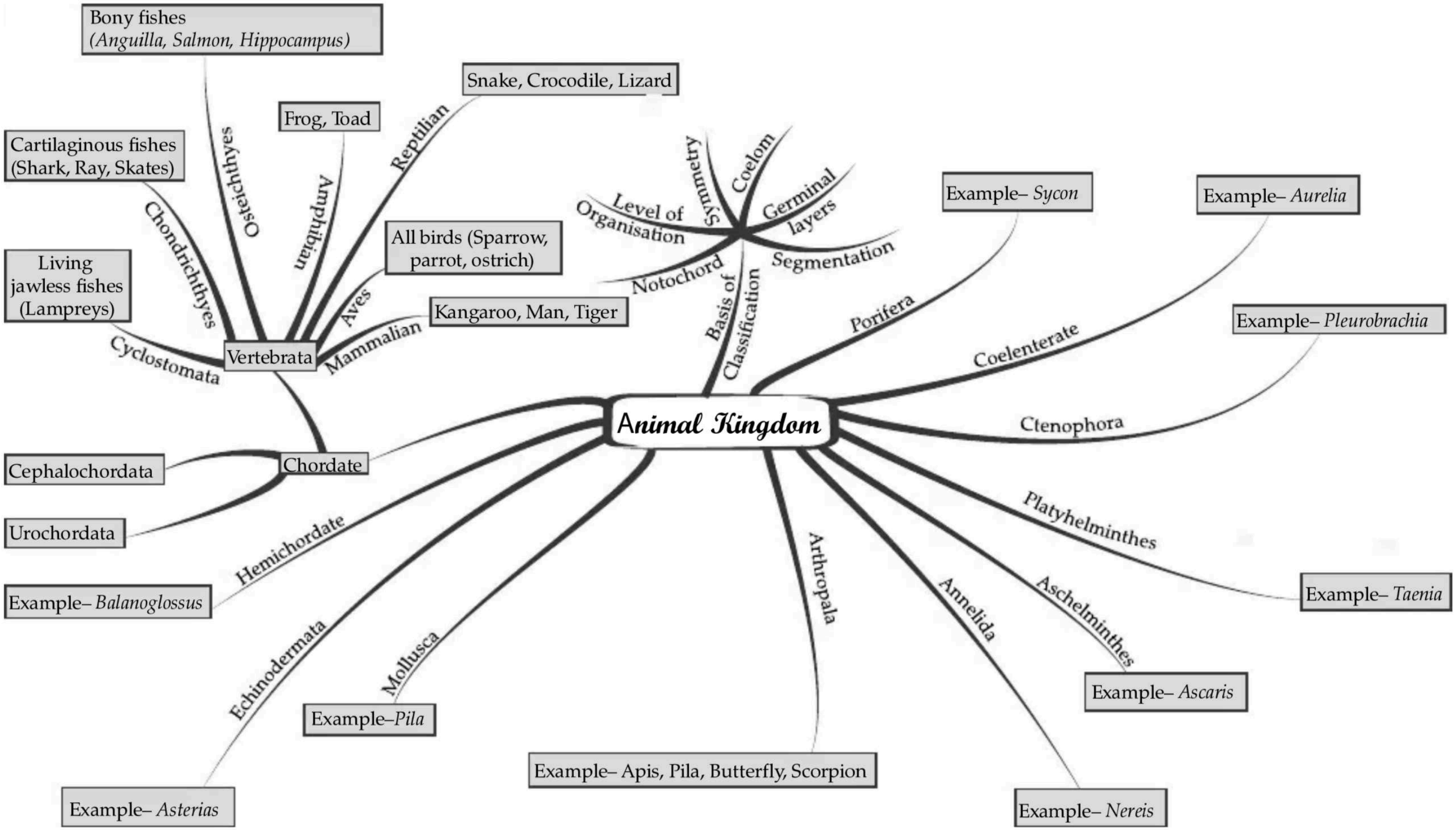


Characters	Five Kingdoms				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present (without cellulose)	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multicellular/ loose tissue	Tissue/ organ	Tissue/organ/ organ system
Mode of nutrition	Autotrophic (chemosynthetic and photosynthetic) and Heterotrophic (saprophytic/parasitic)	Autotrophic (Photosynthetic) and Heterotrophic	Heterotrophic (Saprophytic/ Parasitic)	Autotrophic (Photosynthetic)	Heterotrophic (Holozoic/ Saprophytic etc.)

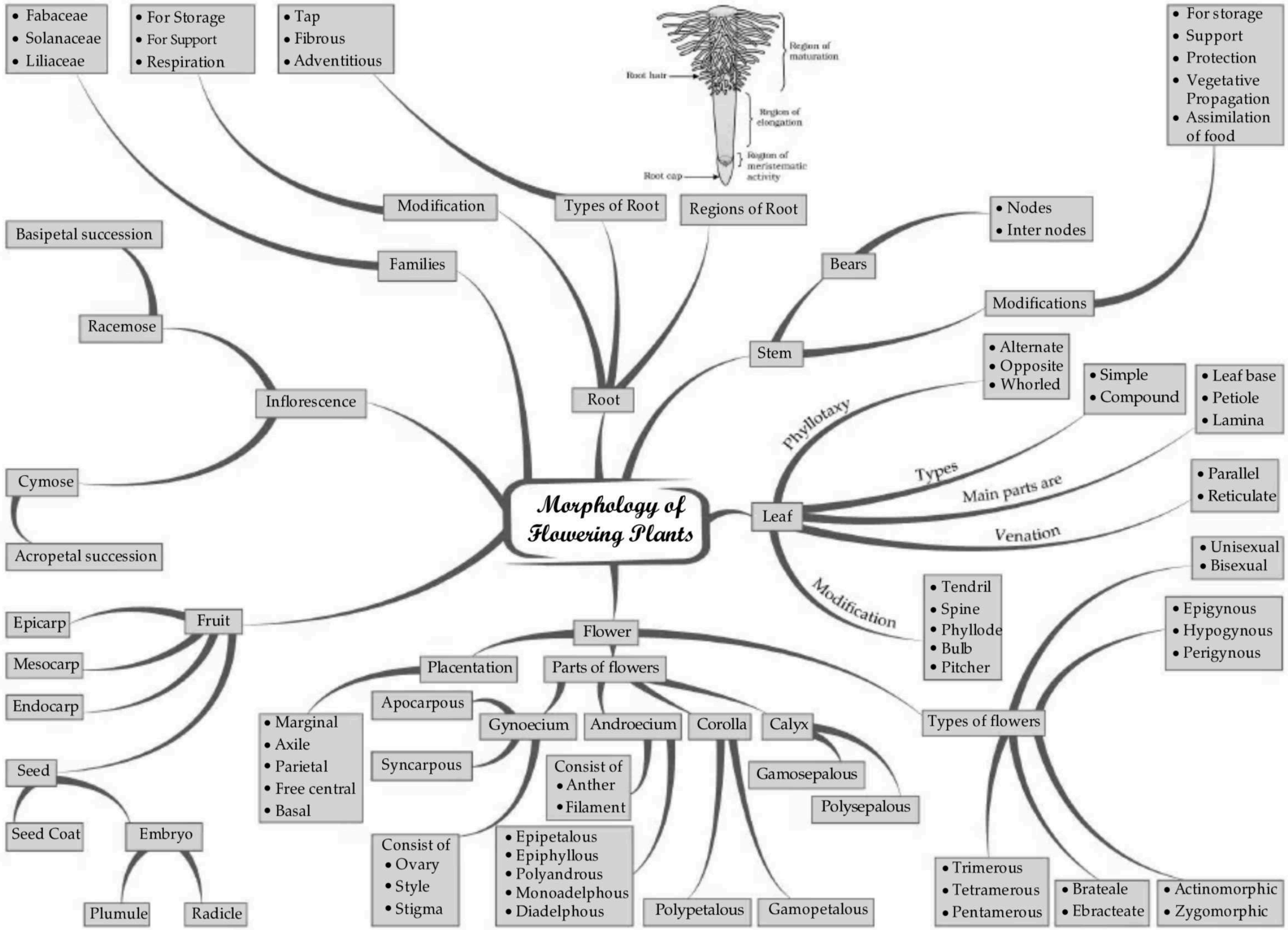
Classes	Common Name	Major Pigments	Stored Food	Cell Wall	Flagellar Number and Position of Insertions	Habitat
Chlorophyceae	Green algae	Chlorophyll *a, b*	Starch	Cellulose	2-8, equal, apical	Fresh water, brackish water, salt water
Phaeophyceae	Brown algae	Chlorophyll *a, c,* fucoxanthin	Mannitol, laminarin	Cellulose and algin	2, unequal, lateral	Fresh water (rare) brackish water, salt water
Rhodophyceae	Red algae	Chlorophyll *a, d,* phycoerythrin	Floridean starch	Cellulose, pectin and poly sulphate esters	Absent	Fresh water (some), brackish water, salt water (most)



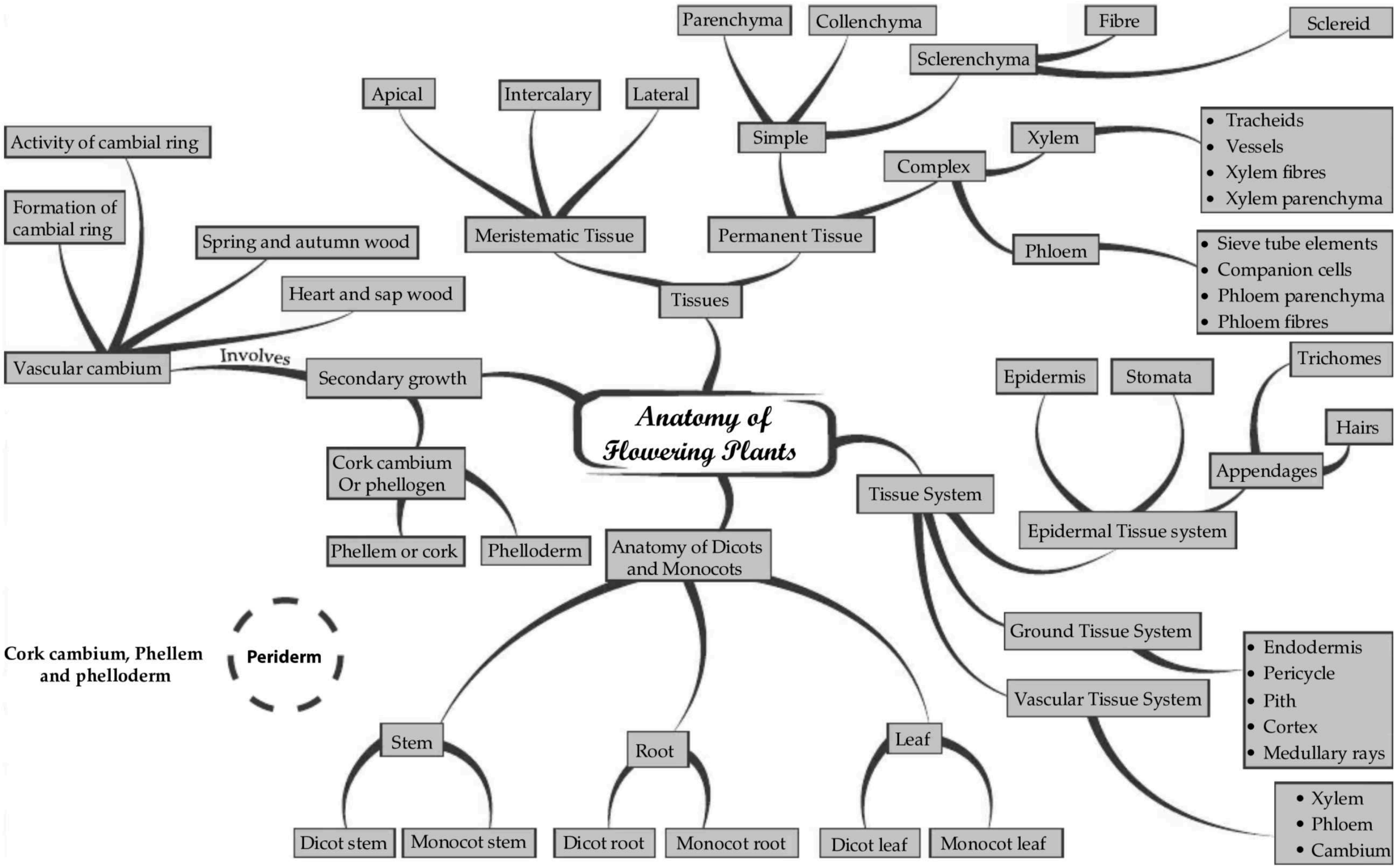
CHAPTER : 4 ANIMAL KINGDOM



CHAPTER : 5 MORPHOLOGY OF FLOWERING PLANTS



CHAPTER : 6 ANATOMY OF FLOWERING PLANTS

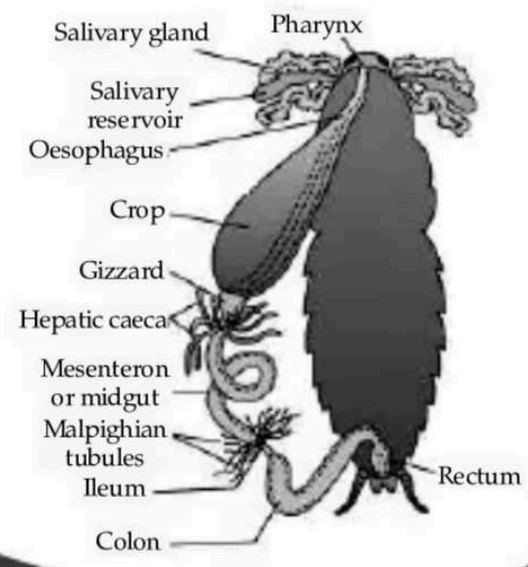


CHAPTER : 7 STRUCTURAL ORGANIZATION IN ANIMALS

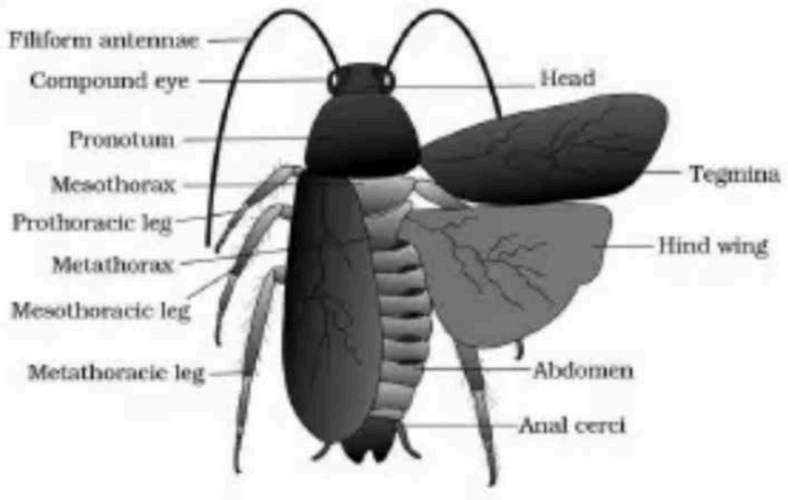
Structural Organization in Animals

Cockroach (*Periplaneta Americana*)

- **Circulatory**
  - Open type- Have poorly developed blood vessels
  - Chambers of heart
  - Alary muscle
  - Anterior aorta
- **Digestive**
  - Foregut
  - Midgut
  - Hindgut
- **Respiratory**
  - Trachea
  - Spiracles
- **Nervous**
  - Segmentally arranged ganglia joined by paired longitudinal connectives
- **Reproduction**
  - Testes
  - Male
  - Ovaries
  - Female
- **Excretion**
  - Malpighian tubules

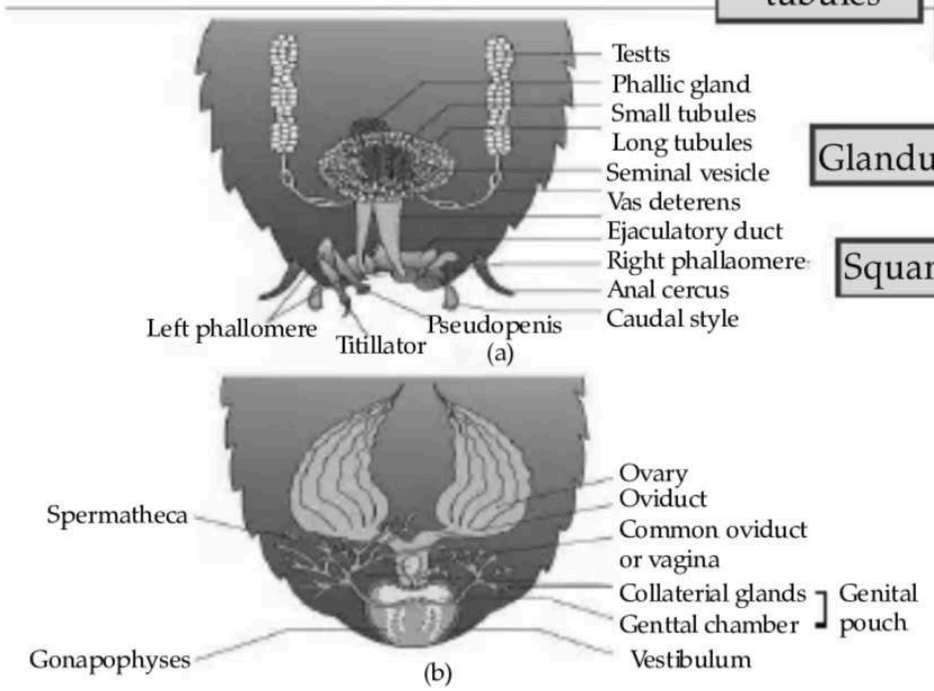


- **Morphology**
  - Head
  - Thorax
    - Prothorax
    - Mesothorax
    - Metathorax
  - Abdomen

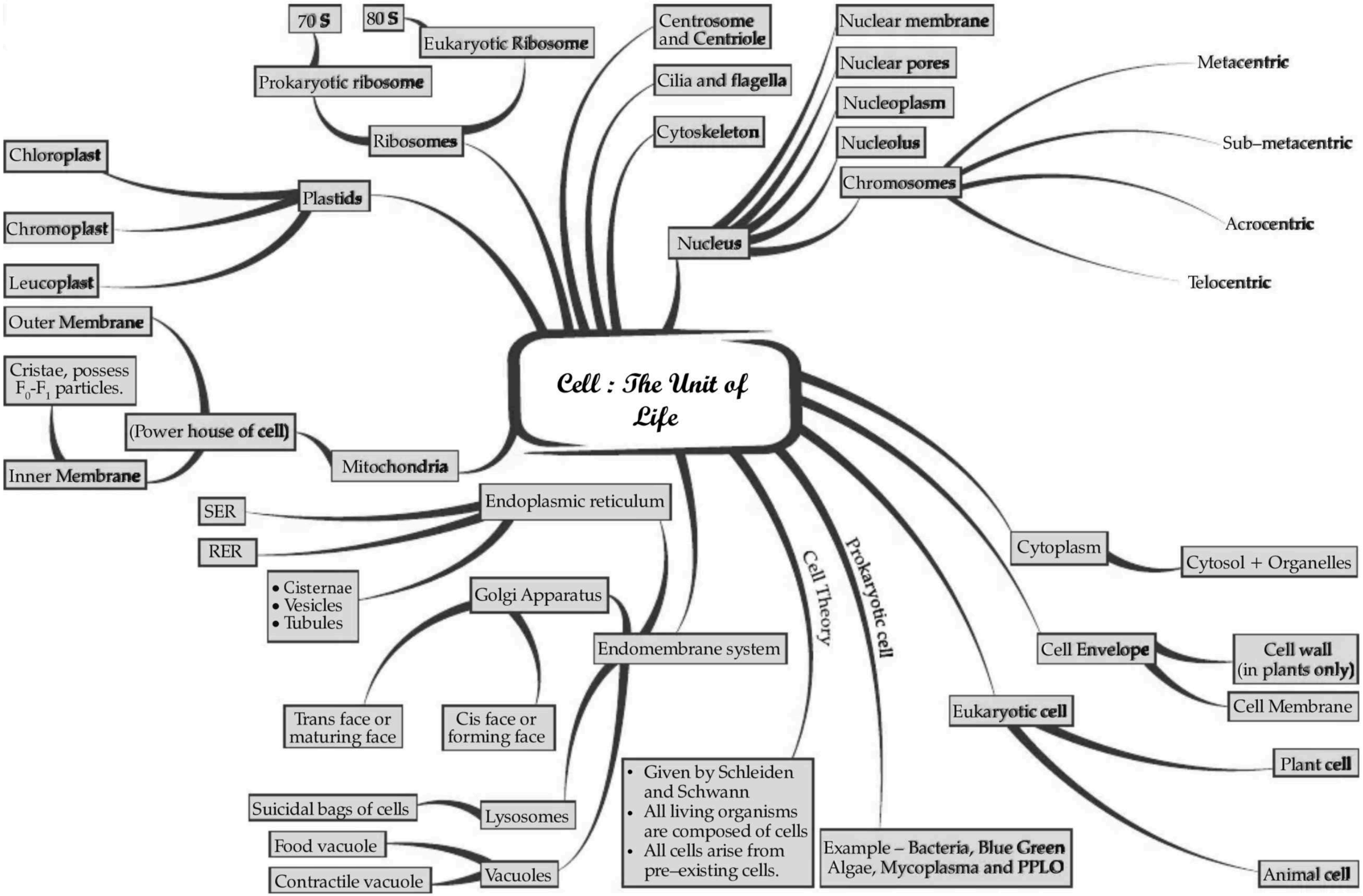


Morphology of Cockroach

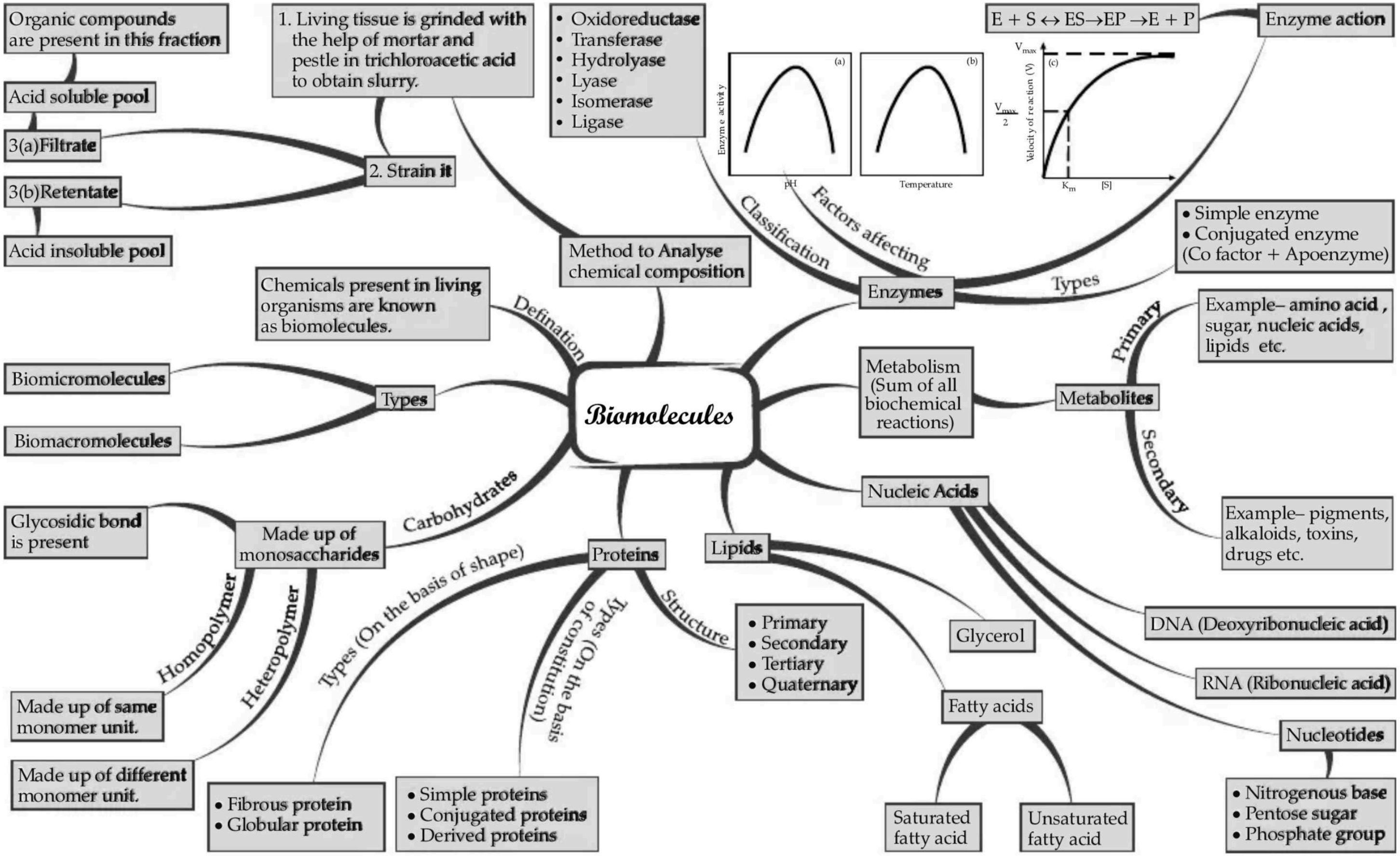
- **Tight**
- **Junctions**
  - Adhering
  - Gap
- **Animal Tissues**
  - **Epithelial**
    - Simple
      - Glandular
      - Squamous
      - Cuboidal
      - Columnar
      - Ciliated
    - Compound
  - **Muscular**
    - Skeletal (Striated)
    - Smooth (Non-striated)
    - Cardiac
  - **Connective**
    - Fluid connective tissue.
      - Lymph
      - Blood
    - Loose connective tissue
      - Areolar
      - Adipose
    - Dense connective tissue
      - Dense Regular
      - Dense irregular
    - Supportive connective tissue
      - Cartilage
      - Bone
  - **Neural**
    - Neuron
      - Cell body
      - Dendrons
      - Axon
    - Neuroglial



CHAPTER : 8 CELL : THE UNIT OF LIFE



CHAPTER : 9 BIOMOLECULES

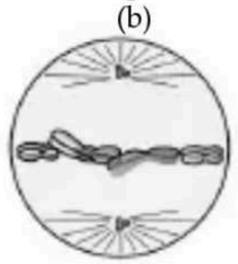


CHAPTER : 10 CELL DIVISION AND CELL CYCLE


Early Prophase

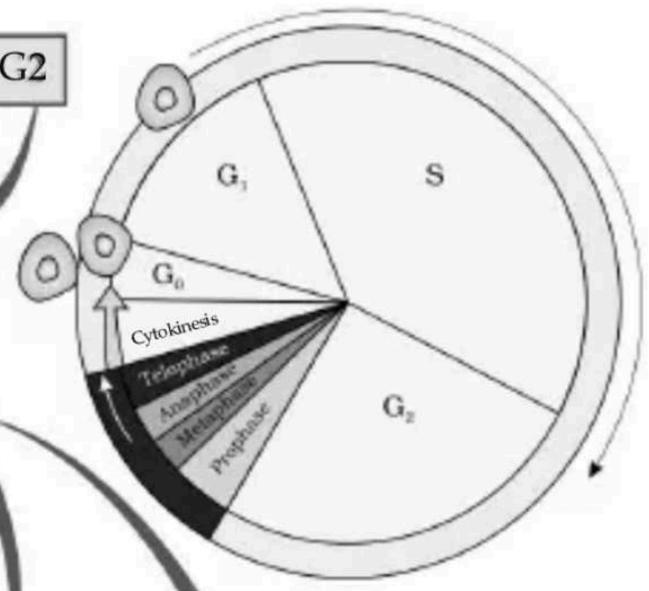

Late Prophase
(a)


Metaphase
(b)


Stages of Mitosis

Cell Division and Cell Cycle

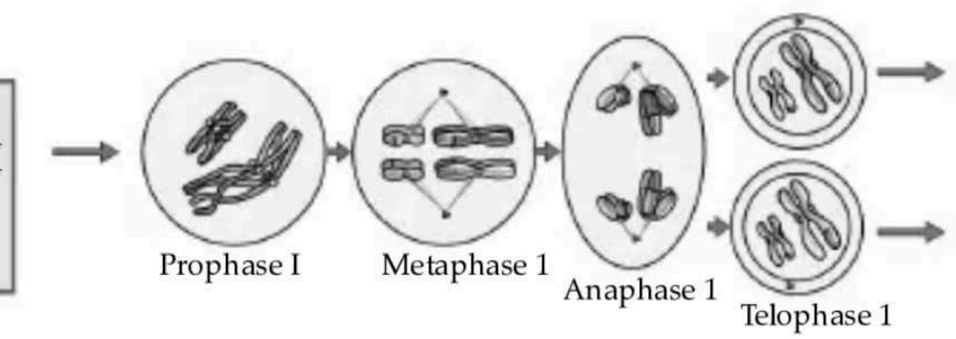
- Cell Cycle
  - G1
  - S
  - G2
  - Inter phase
  - Interphase
  - M–phase
    - Karyokinesis
      - Prophase
      - Metaphase
      - Anaphase
      - Telophase
    - Cytokinesis
      - Animal cell
        - Starts with appearance of furrow in the plasma membrane
      - Plant cell
        - Cell wall formation starts in the centre

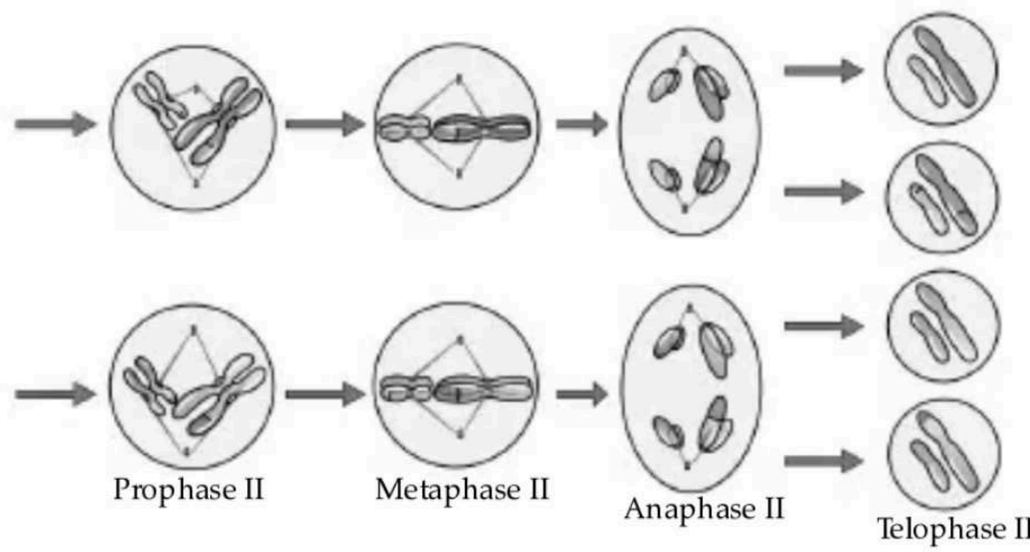


- Mitosis- Equational Division



- Meiosis- Reductional Division
  - Meiosis I
    - Prophase I
      1. Leptotene (Duplicated chromosomes start to condense)
      2. Zygotene (Synapsis begins)
      3. Pachytene (Synapsis complete, Crossing over occurs)
      4. Diplotene (Synaptonemal complex starts disappearing, Bchiasma visible)
      5. Diakinesis (Bivalent ready for metaphase)
    - Metaphase I
    - Anaphase I
    - Telophase I
  - Meiosis II
    - Prophase II
    - Metaphase II
    - Anaphase II
    - Telophase II





CHAPTER : 11 TRANSPORT IN PLANTS

Transport in Plants

Means of Transport

- One molecule move independent of other.
- When both molecules move in the same direction
- Both molecules move in opposite direction

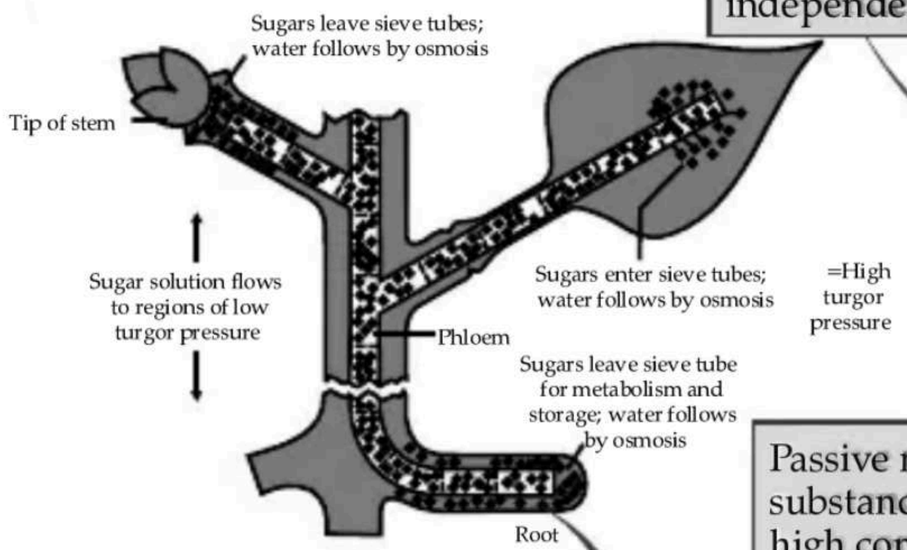
Uniport, Symport, Antiport → Facilitated Diffusion

Diffusion → Passive movement of substances from high concentration to low concentration.

Active Transport → Energy is required to pump molecules against concentration gradient.

Plant Water Relations

- Water Potential (ψ_w) : $\Psi s + \Psi p$
- Osmosis Diffusion of solvent (water) across the semi-permeable membrane.
- Plasmolysis: Shrinkage of protoplasm in a cell, away from its cell wall and toward the center
- Imbibitions



Mass flow hypothesis

It is the most accepted theory for the translocation of sugar from source to sink.
- Source cells are cells which are the supplier of the food material, mainly leaves.
- Sink cells are cells which require food material for growth and repair.

Transpiration

It is the loss of water through stomata of leaves and other aerial parts of plants in form of water vapour.

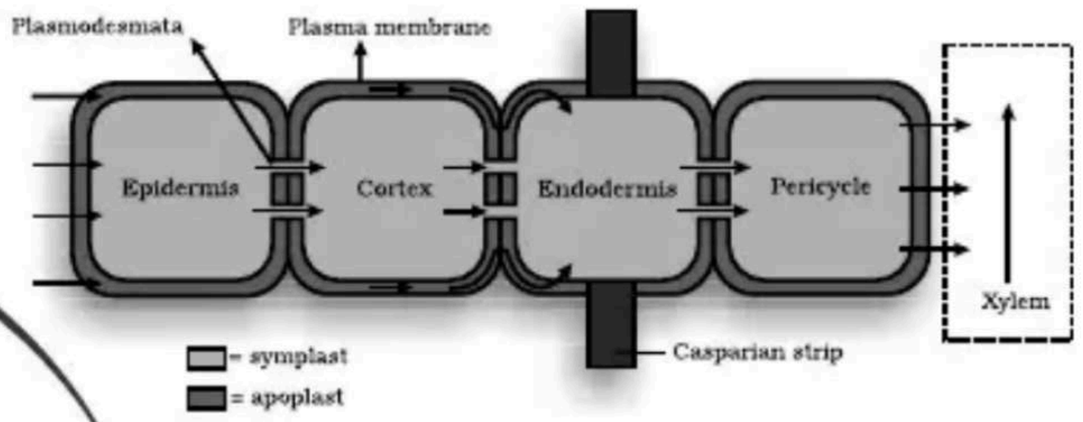
Water movement up a plant

- Root pressure → It is a positive pressure that develops in the roots by the active absorption of nutrients from the soil.
- Transpiration pull → Transpiration creates transpirational pull for absorption and transport in plants.

Uptake and Transport of Mineral Nutrients

- Uptake of Mineral Ions → Minerals are absorbed by the roots by passive and active transport. The active uptake of ions requires ATP energy.
- Translocation of Mineral Ions → Unloading of mineral ions occur at fine vein endings of the leaves through diffusion. Some minerals are also remobilised from old senescing parts N, P K, S.

Absorption of water

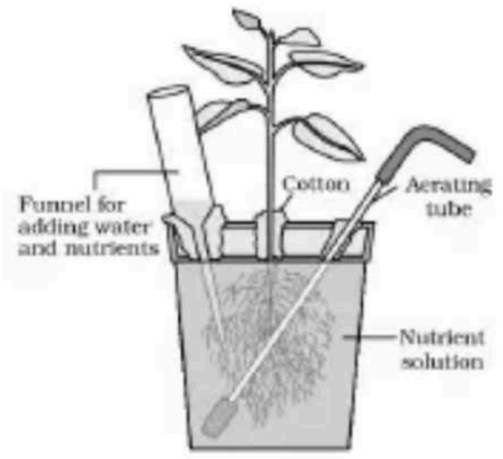


- Symplast Pathway
- Apoplast Pathway

CHAPTER : 12 MINERAL NUTRITION

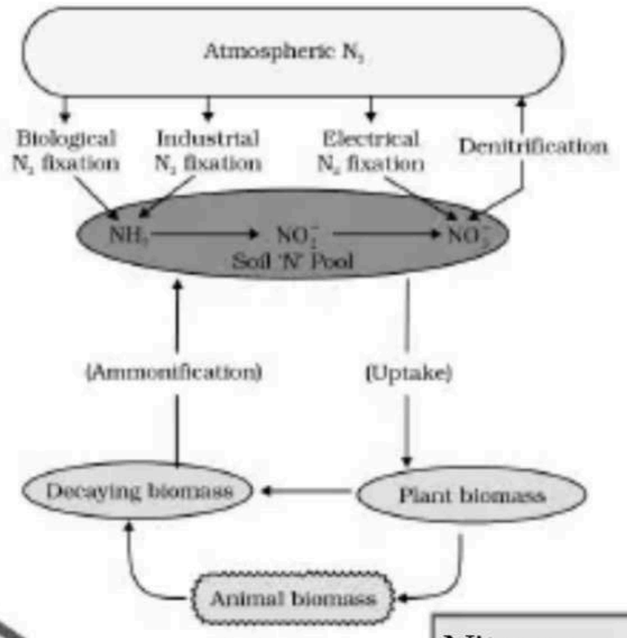
Mineral Nutrition

Methods to Study the Mineral Requirements of Plants



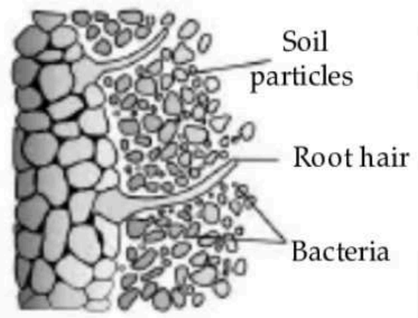
Hydroponics

Plant is cultured in soil–free defined minerals solution

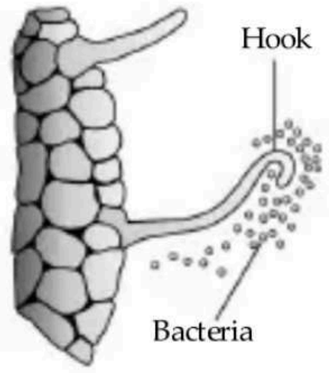


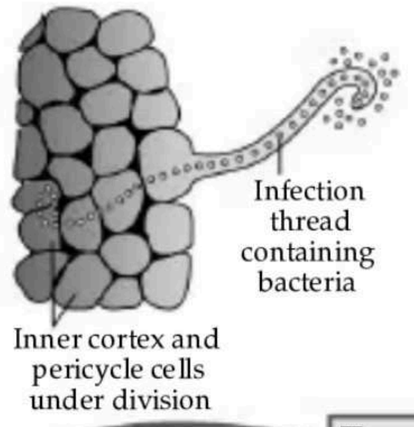
Nitrogen cycle

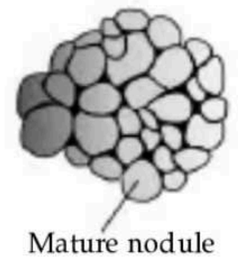
Metabolism of Nitrogen



(a)



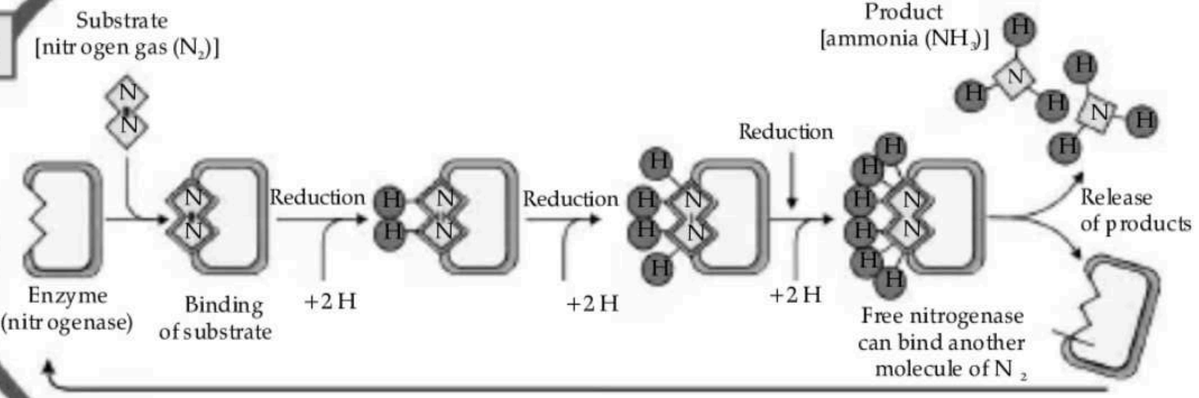




Development of root nodules in Soyabean

Fate of ammonia

- Transamination
- Reductive amination



Steps of conversion of atmospheric nitrogen to ammonia by nitrogenase enzyme complex found in nitrogen-fixing bacteria

Essential Mineral Elements

Toxicity of Micronutrients

In higher dose, micronutrient become toxic. Toxicity of one element may lead to deficiency of other elements since the former may inhibit the uptake of latter.

Criteria

- Element must be absolutely necessary for normal growth and development
- Element must be directly involved in the metabolism of plant.
- Requirement of element must be specific and not replaceable

Two Categories of Elements

Macronutrient

Required in large quantity. E.g. Hydrogen, Oxygen, Nitrogen, phosphorus, sulphur, potassium, calcium and Magnesium

Micronutrient

Required in less quantity. E.g. Cobalt, manganese, Zinc, Iron, Copper, Molybdenum, Boron, chlorine and nickel.

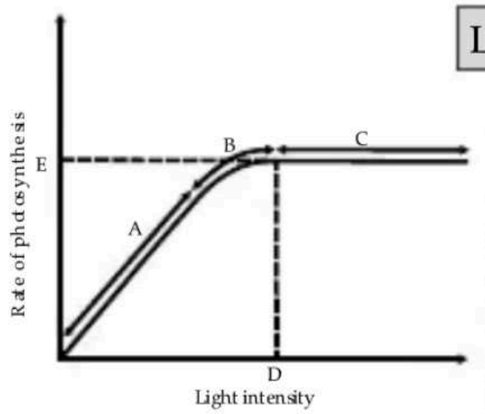
Deficiency Symptoms

Deficiency symptoms are morphological changes in plants, indicating nutrient deficiency.

Mechanism of Absorption of Elements

- Apoplast
- Symplast

CHAPTER : 13 PHOTOSYNTHESIS IN HIGHER PLANTS



Photosynthesis in Higher Plants

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{light}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

Photosynthesis an enzyme regulated anabolic process of manufacture of organic compounds inside the chlorophyll containing cells, from carbon oxide and water with the help of sunlight as source of energy.

Factors Affecting Photosynthesis

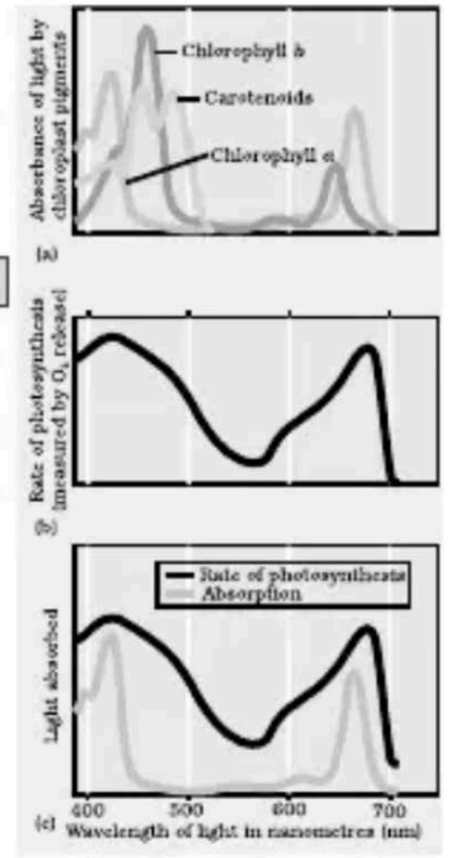
- Light
- Carbon dioxide
- Temperature
- Water

Photosynthesis as a Means of Autotrophic Nutrition

Site of photosynthesis

Chloroplast in leaves

- light Reaction — Grana
- Dark Reaction — Stroma

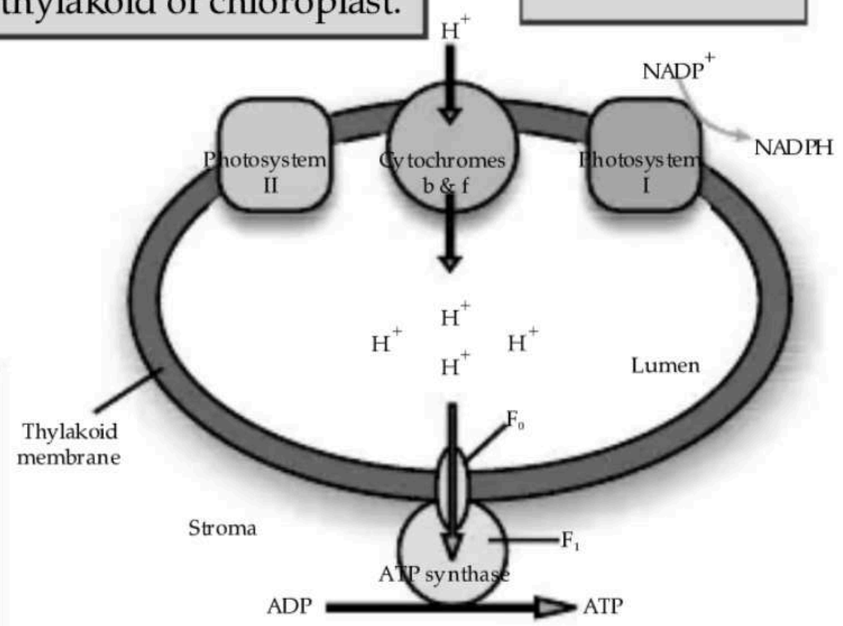


Pigments Involved in Photosynthesis

- Chl a
- Chl b
- Xanthophyll
- Carotenoids

Chemiosmotic hypothesis

Explains the mechanism of ATP synthesis in thylakoid of chloroplast.



Photorespiration

- It is the light induced respiration in green plants.
- It is a process in which there is no formation of ATP or NADPH, but there is utilization of ATP with release of CO_2.

Light Reaction

Photophosphorylation

- Cyclic
  - Involves only photosystem I.
  - Only ATP is formed.
  - Electrons move in a closed circle.
- Non-cyclic
  - Involves both photosystem I and II.
  - Both ATP and NADPH are produced.
  - Electrons do not move in a closed circle.

Biosynthetic Phase Or Dark reaction

- C4 or Hatch and Slack pathway
  - Occurs in mesophyll and bundle sheath cells.
  - Enzyme RuBisCo is absent from the mesophyll cells
- C3 or Calvin cycle
  - Carboxylation
  - Reduction
  - Regeneration
  - For every CO_2 molecules, 3 molecules of ATP and 2 molecules of NADPH are required.

CHAPTER : 14 RESPIRATION IN PLANTS

Respiration in Plants

Definition

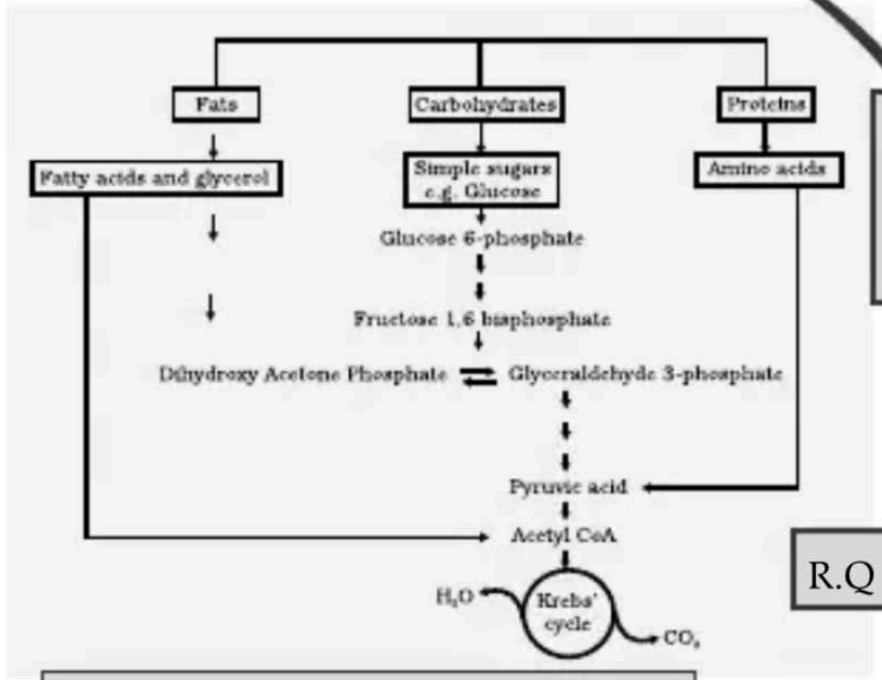
Respiration is an energy releasing enzymatically controlled catabolic process which involves a step–wise oxidative breakdown of food substances inside living cells.

Repiratory Quotient.

$$RQ = \frac{\text{Volume of } CO_2 \text{ Evolved}}{\text{Volume of } O_2 \text{ Consumed}}$$

R.Q. values = 1 (for carbohydrate...
R.Q.= 0.7...
R.Q. = 0.9....

R.Q formula

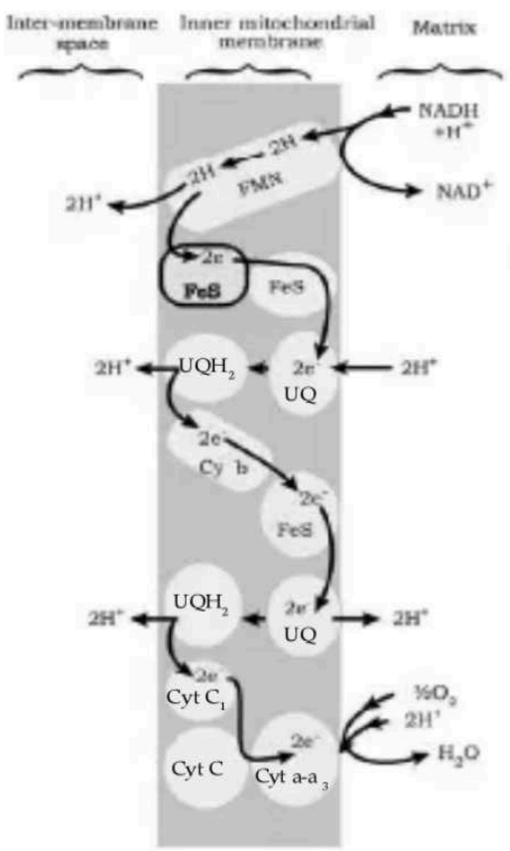


Amphibolic Pathway

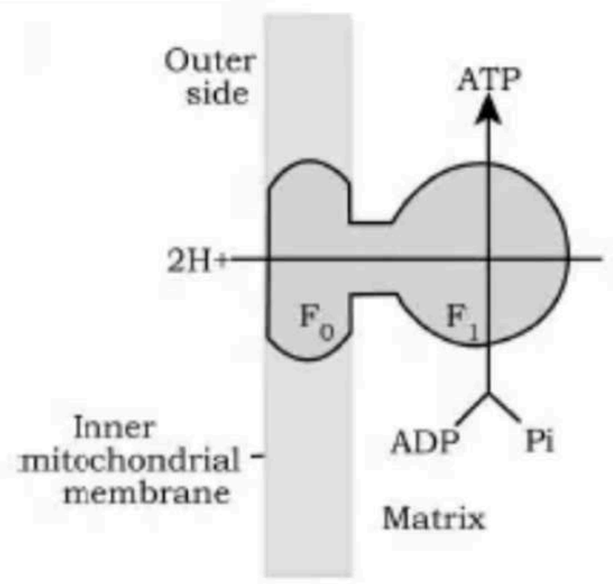
Respiration is termed as amphibolic pathway as it involves both anabolism and catabolism.

Aerobic respiration

TCA Tricarboxylic Acid (Matrix of mitochondria)

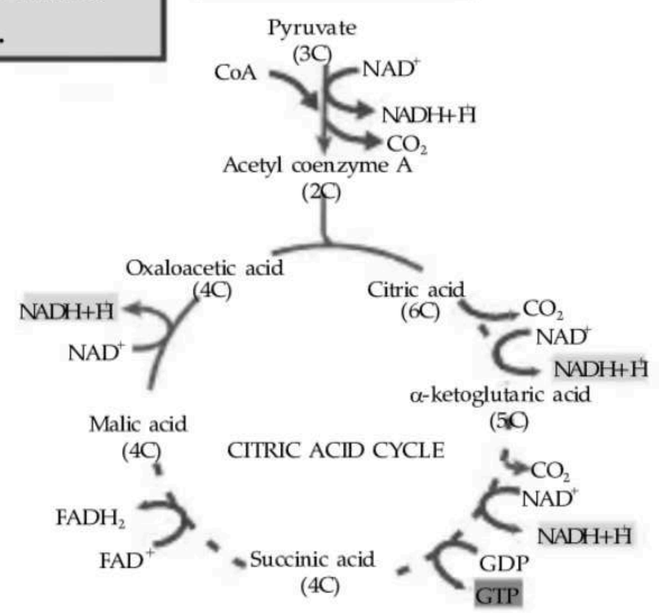


2nd of - Electron transport chain (Site: Inner mitochondrial membrane)



1st of - Oxidative Phosphorylation Site: F0- F1 particles in the inner mitochondrial membrane.

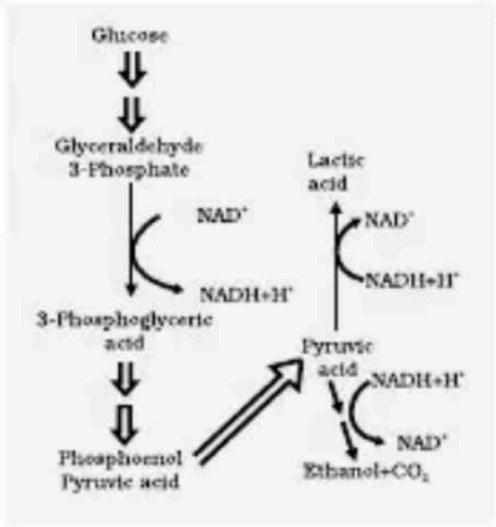
3rd of- TCA/ Tricarboxylic acid (Site: Matrix of mitochondria)



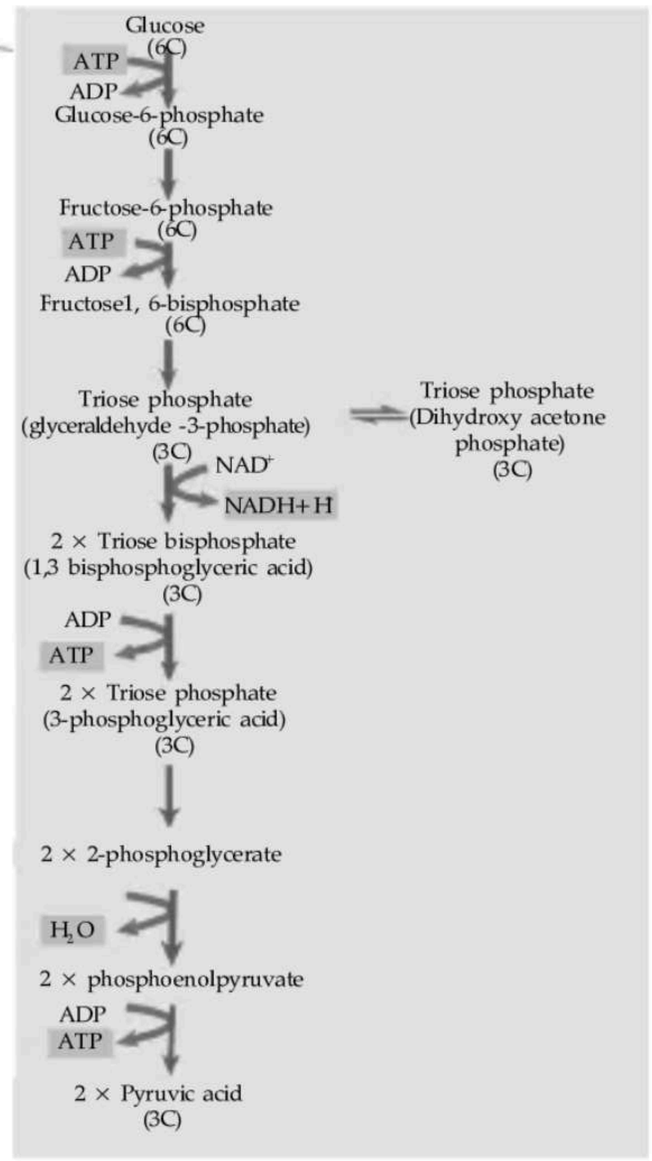
The Citric acid cycle

Glycolysis box

Glycolysis Or EMP pathway (Site: Cytoplasm)



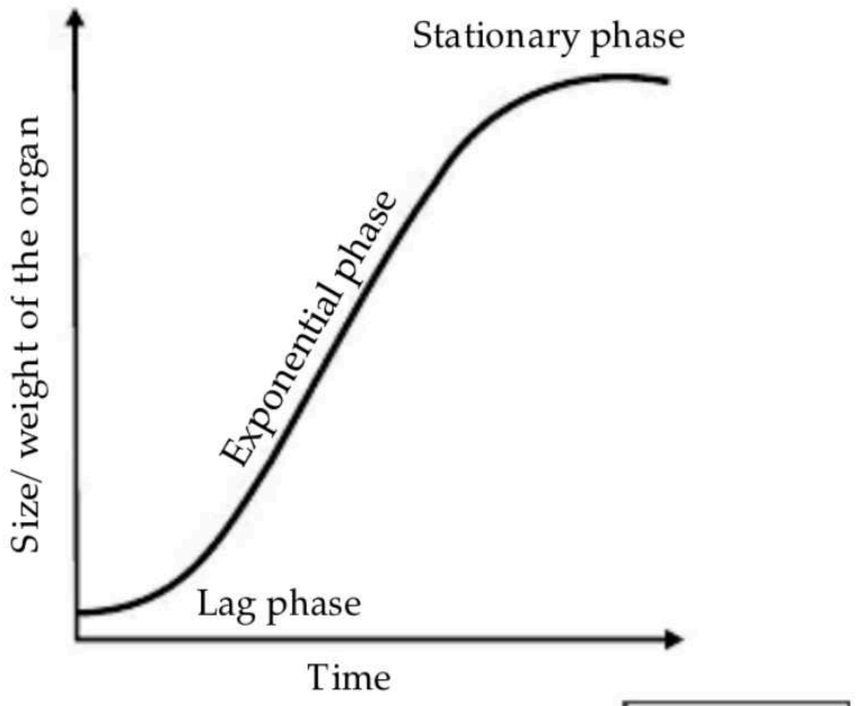
Fermentation (Incomplete oxidation of pyruvic acid, under anaerobic condition)



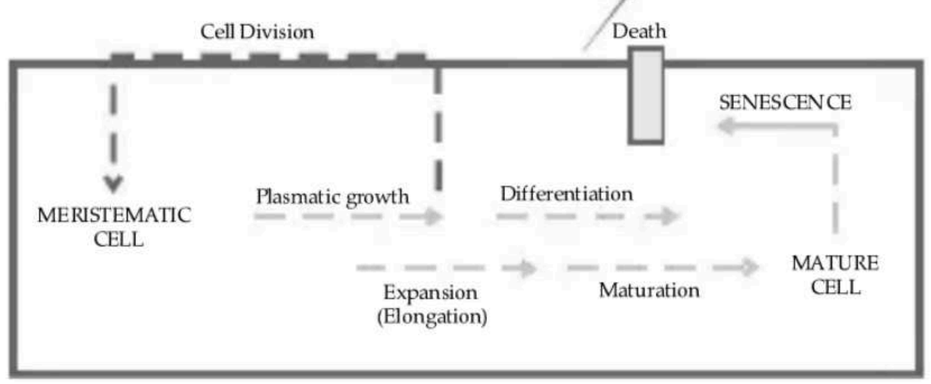
CHAPTER : 15 PLANT GROWTH AND DEVELOPMENT

Plant Growth and Development

- Photoperiodism
  - Short day plant — Example– *Chrysanthemum*
  - Long day plant — Example– Radish, barley
  - Day neutral plant — Example– Tomato
- Seed germination
  - Root
  - Shoot
  - Radicle
  - Plumule
- Phases of plant growth
  - Meristematic
  - Elongation
  - Maturation

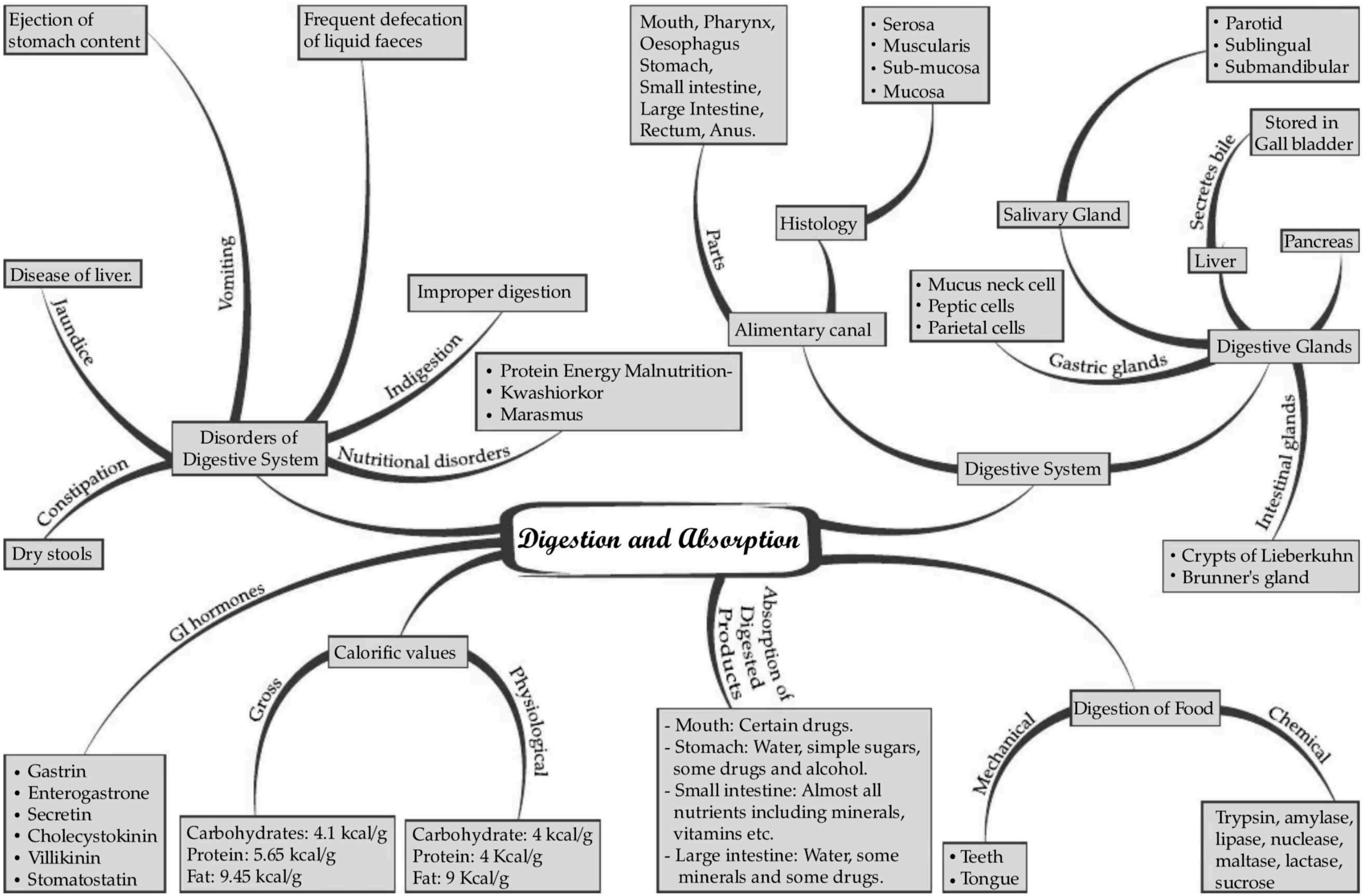


- Quantitative comparison between growth is done by
  - Absolute growth rate — Measurement and comparison of total growth per unit time.
  - Relative growth rate — Growth of a particular system per unit time, expressed on a common basis.
- Growth rate — Increase in growth per unit time.
  - Arithmetic $L_t = L_o + rt$
  - Geometric $W_1 = W_0 e^{rt}$
- Vernalisation — Promotion of flowering by a period of low temperature.
- Plant growth regulators (Phytohormones)
  - Auxin
  - Gibberellin
  - Cytokinin
  - Ethylene
  - ABA
- Conditions of growth
  - Water
  - Oxygen
  - Nutrients
  - Optimum temperature
  - Light
  - Gravity
- Sequence of developmental processes



- Redifferentiation — Process in which dedifferentiated cells becomes mature again and lose their capacity to divide.
- Dedifferentiation — Process in which, the living differentiated cells that by now have lost the capacity to divide, regain their capacity to divide under certain conditions.
- Differentiation — Process in which cells produced by apical meristem become specialized to perform specific function

CHAPTER : 16 DIGESTION AND ABSORPTION



CHAPTER : 17 BREATHING AND EXCHANGE OF GASES

Breathing and Exchange of Gases

Respiratory organs in animals

- Body surface
  - Examples – sponges, coelenterate, flatworm
- Tracheal tube
  - Examples – Insects, centipede, millipede etc.
- Gills
  - Example – Fishes, tadpoles, prawn etc.
- Lungs
  - Example – Terrestrial animals like human being

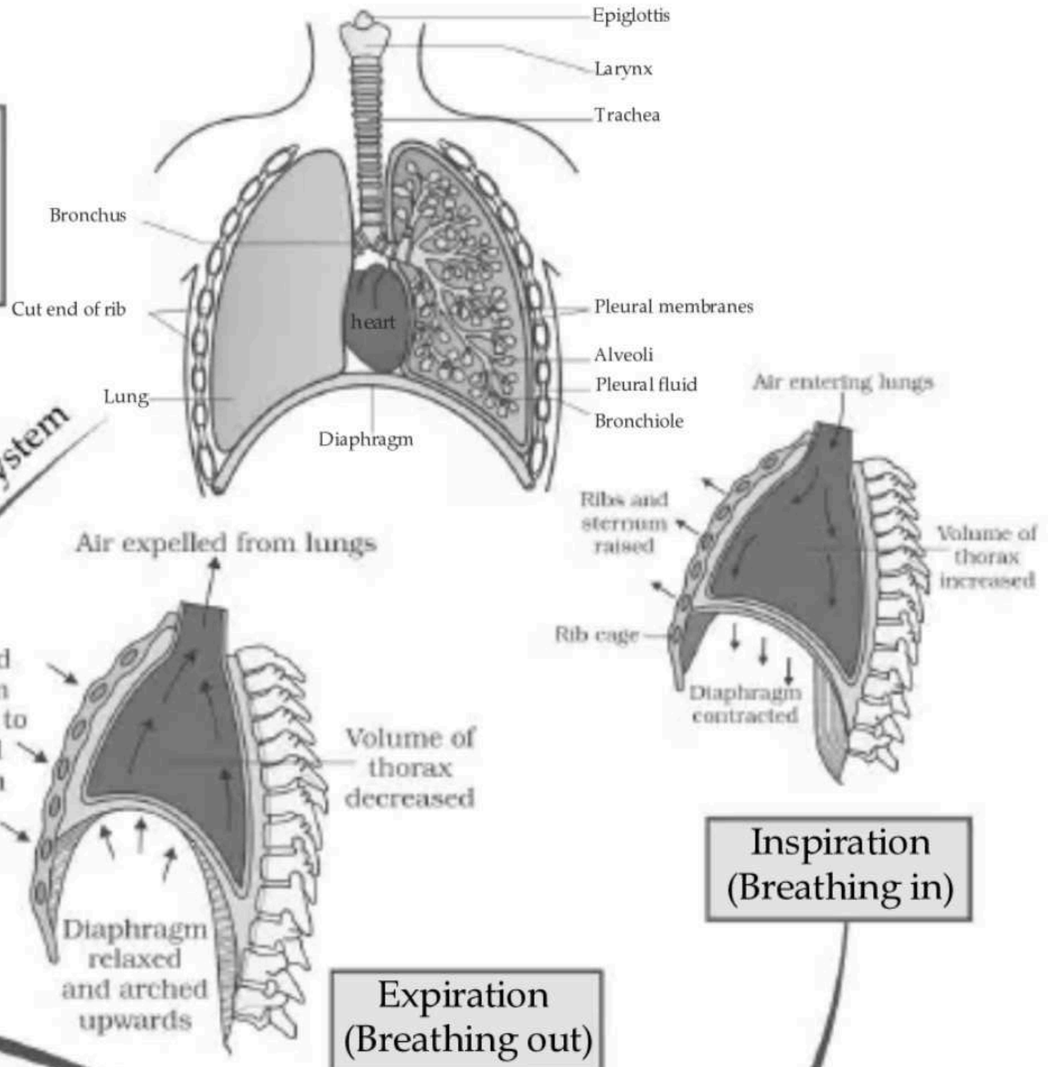
Respiratory disorders

- Occupational Respiratory Disorders
- Emphysema
- Asthma
- Bronchitis
- Pneumonia
- Lung cancer

Regulation of Respiration

- Neural regulation- Involves respiratory centers, afferent and efferent nerves.
- Chemical regulation- Through chemoreceptors.

Human respiratory System



Inspiration (Breathing in)

Expiration (Breathing out)

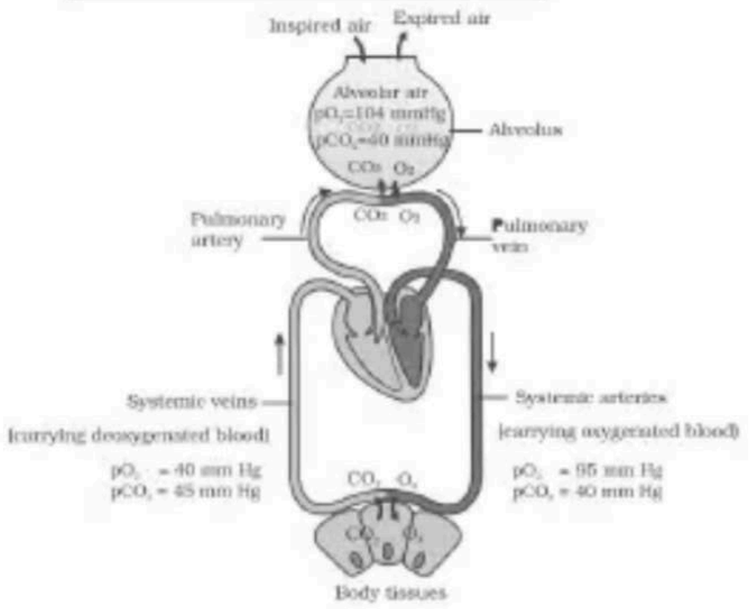
Mechanism of Breathing

Respiratory volumes and capacities

- Tidal volume → 500 ml
- IRV → 2500 ml to 3000 ml
- ERV → 1000 ml to 1100 ml
- RV → 1100 ml to 1200 ml
- IC → TV + IRV
- EC → TV + ERV
- FRC → ERV + RV
- VC → ERV + TV + IRV
- Total lung capacity
  → RV + ERV + TV + IRV or V.C + TRV

Exchange of Gases

Between Alveoli and blood



Between blood and tissues

Transport of Gases

- Transport of O_2
  - As blood plasma (About 3 %)
  - As oxyhaemoglobin (About 97%)
- Transport of CO_2
  - As Carbonic acid (About 7 %)
  - As carbamino- haemoglobin (About 20-25 %)
  - As bicarbonates (About 70%)

$$CO_2 + H_2O \underset{\text{Anhydrase}}{\overset{\text{Carbonic}}{\rightleftharpoons}} H_2CO_3 \underset{\text{Anhydrase}}{\overset{\text{Carbonic}}{\rightleftharpoons}} HCO_3^- + H^+$$

CHAPTER : 18 BODY FLUIDS AND CIRCULATION

Body Fluids and Circulation

Blood

- Plasma
- Formed Elements
  - Platelets
  - RBC
  - WBC
    - Granulocytes
      - Neutrophils
      - Eosinophils
      - Basophils
    - Agranulocytes.
      - Lymphocytes
      - Monoctyes

Blood Groups

- ABO grouping

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB,A,B,O
O	nil	anti-A,B	O

- Erythroblastosis foetalis
  - Rh + ve
  - Rh – ve

Coagulation of blood

When an injury is caused to a blood vessel, bleeding starts which is stopped by a process called blood clotting.

Lymph (Tissue Fluid)

It is a colourless fluid containing specialized lymphocytes that provide immune response to body.

Blood vessels

- Types
  - Arteries
  - Vein
  - Capillaries
- Histology
  - Tunica interna
  - Tunica media
  - Tunica externa

Evolution of heart

- Fishes- 2 chambered heart
- Amphibians- 3 chambered heart
- Reptiles (except crocodiles)- 3 chambered heart
- Crocodiles, Birds and Mammals- 4 chambered heart.

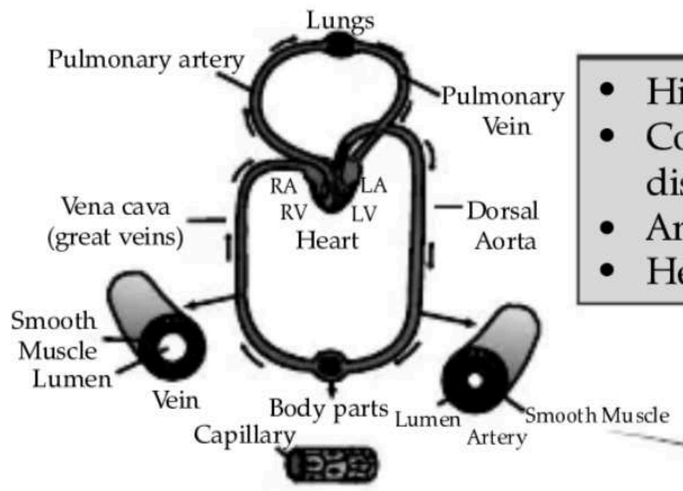
Circulatory pathways

- Open: E.g. Arthropods, Molluscs
- Closed: E.g. Annelids, Echinoderms, Vertebrates

Disorders

- High B.P
- Coronary artery disease
- Angina
- Heart failure

Double Circulation



Regulation of Cardiac Activity

- ANS
  - Sympathetic nerve
  - Parasympathetic nerve
- Adrenal medullar hormone

Human heart

- Heart valves
  - Tricuspid valve
  - Bicuspid valve
  - Semilunar valve
- Histology
  - Endocardium
  - Myocardium
  - Pericardium
- Types
  - Myogenic heart
  - Neurogenic heart
- Cardiac cycle
  - Atrial systole
  - Ventricular systole
  - Cardiac diastole
- Heart Sounds
  - Lubb
  - Dub

CHAPTER : 19 EXCRETORY PRODUCTS AND THEIR ELIMINATION

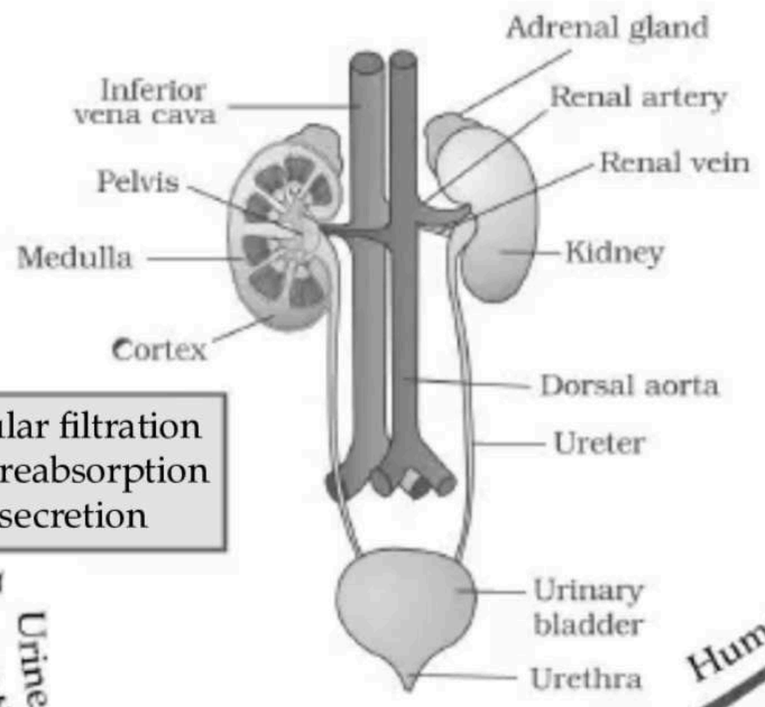
Excretory Products and Their Elimination

Modes of Excretion

- **Ureotelism**
  - Excretory product is Urea
  - Example–mammals, Terrestrial Amphibians
- **Ammonotelism**
  - Excretory product is ammonia
  - Example– bony fishes, aquatic amphibians and insects
- **Uricotelism**
  - Excretory product is uric acid
  - Example- Birds

Urine Formation

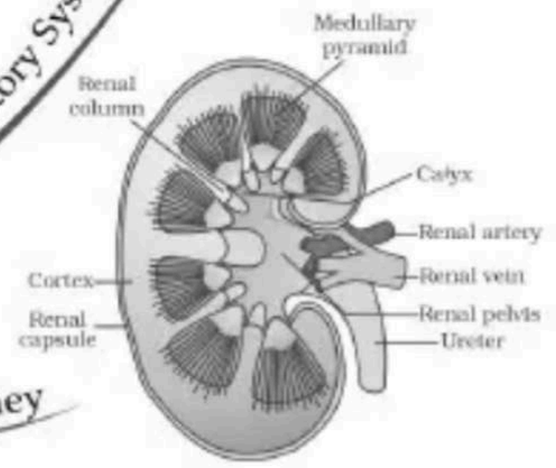
- Glomerular filtration
- Tubular reabsorption
- Tubular secretion



Human Excretory System

- A Pair of Kidneys
- Ureter
- Urinary Bladder
- Urethra

Structure of kidney

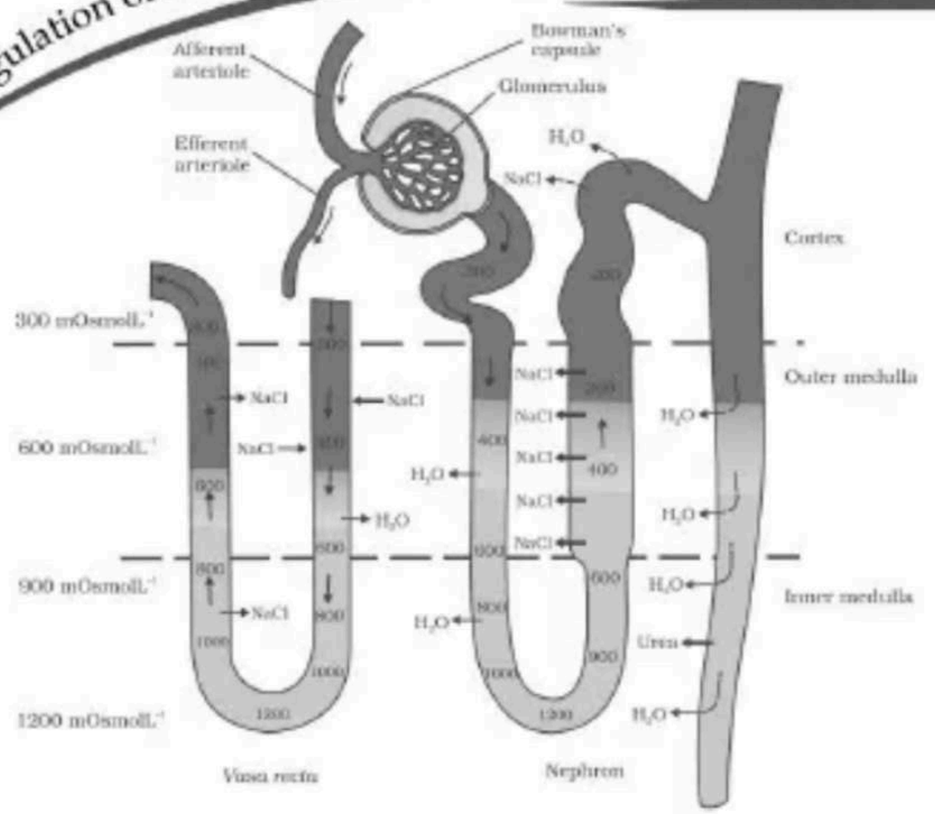


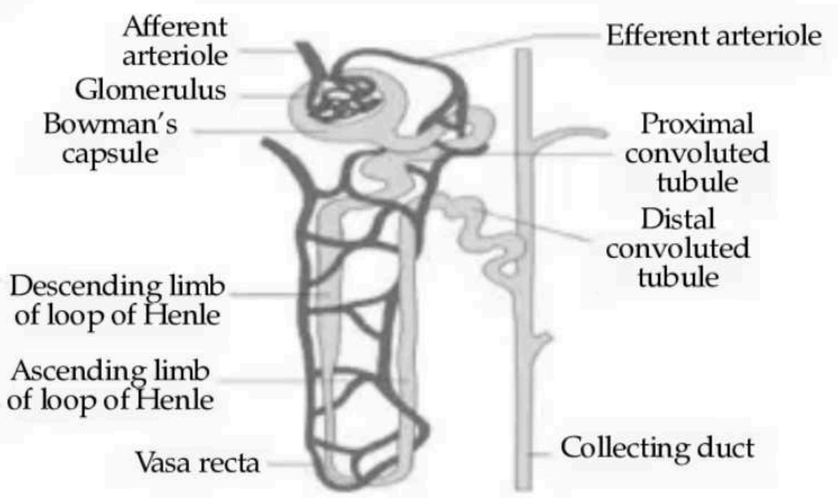
Function of the Tubules

- PCT
  70% of electrolytes are reabsorbed
- Henle's Loop
  Maintains high osmolarity of medullary fluid
- DCT
  Conditional reabsorption of Na^+ and water
- Collecting Duct
  Large amount of water is reabsorbed to produce concentrated urine

Counter-current mechanism

- Henle's loop
- Vasa rectae





Disorders of the excretory System

- Uremia
- Renal failure
- Renal Calculi
- Glomerulonephritis

Micturition

The process of expulsion of urine from urinary bladder

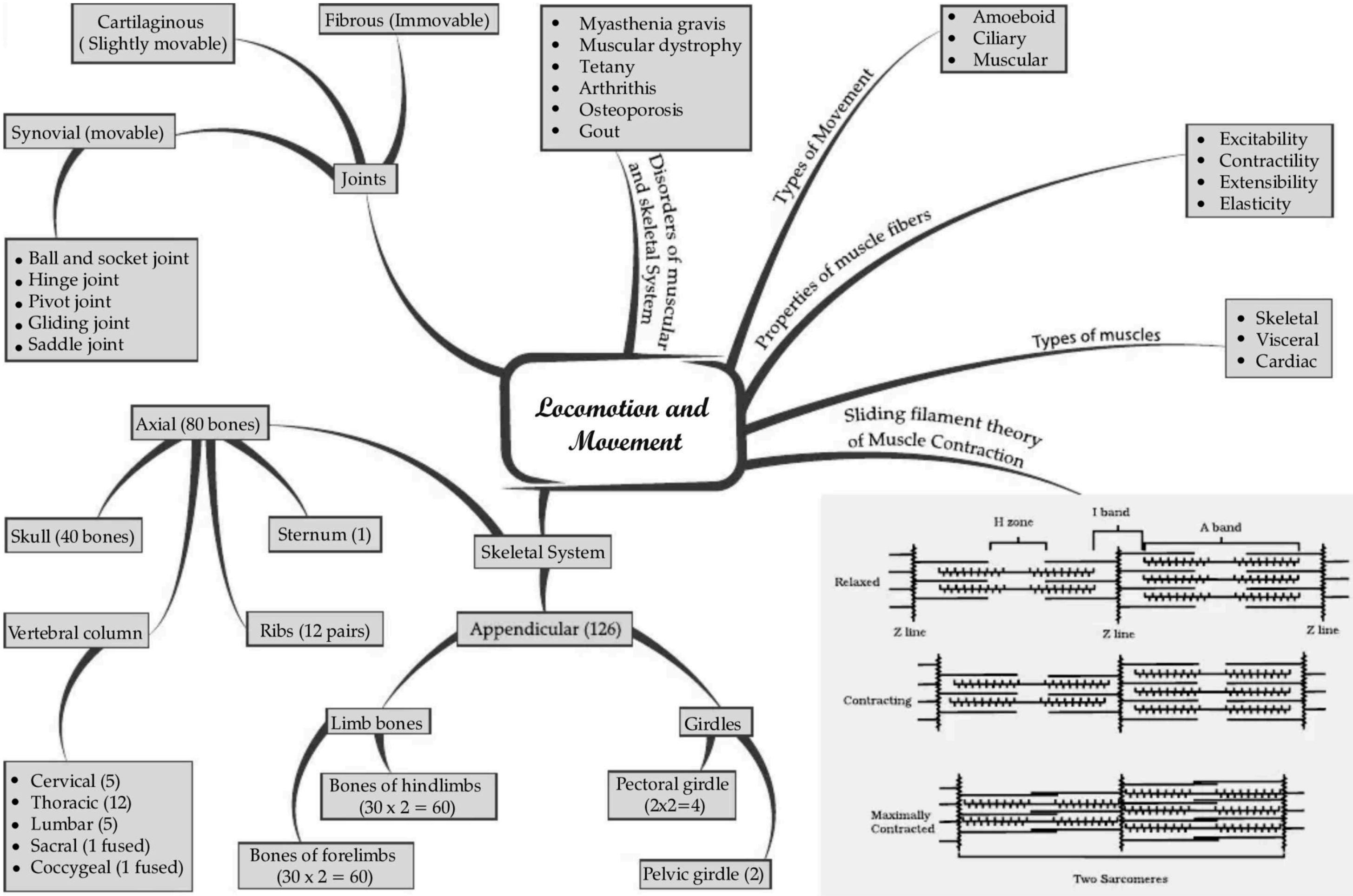
Regulation of kidney

- By ADH (Vasopressin)
- By Juxtaglomerular Apparatus (Renin-Angiotensin mechanism)
- By ANF (in heart)

Other organs in excretion

- Liver
- Lungs
- Skin

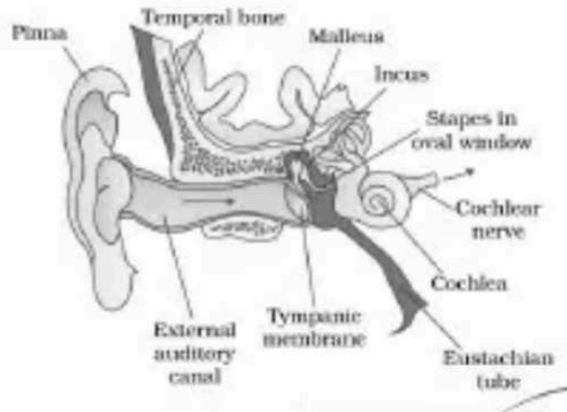
CHAPTER : 20 LOCOMOTION AND MOVEMENT



CHAPTER : 21 NEURAL CONTROL AND COORDINATION

Neural Control and Coordination

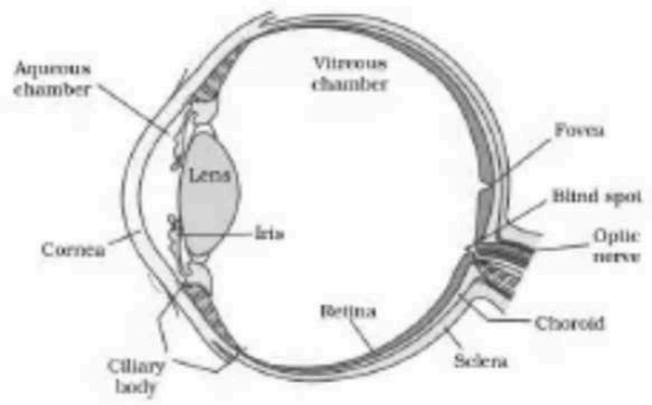
- Neural organization is very simple
- Brain is present along with number of ganglia and neural tissues
- More developed neural system
- Lower invertebrates
- Insects
- Vertebrates
- Neural System
- Sense organs
- Human Ear
- Human Eye



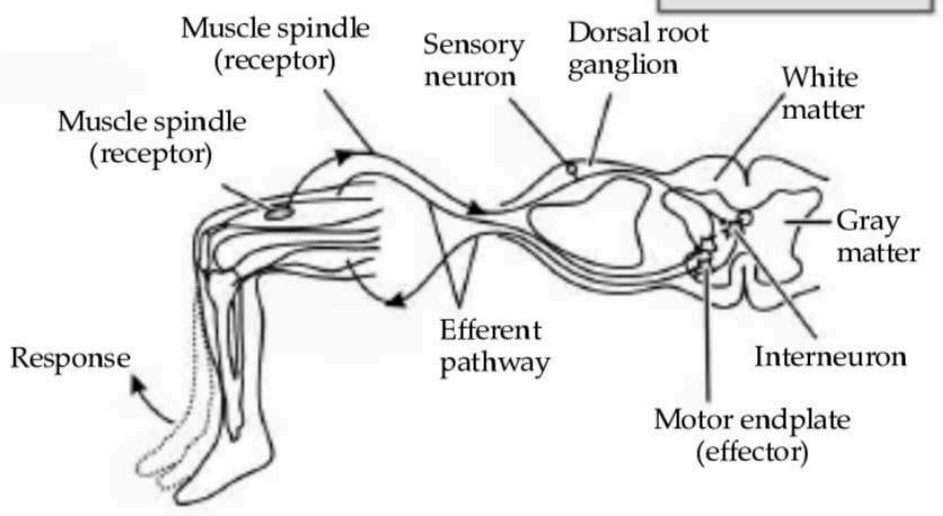
- Outer Ear : Pinna, Ear canal, Tympanic membrane
- Middle Ear - Three ear ossicles
- Inner Ear - Utriculus, Sacculus and Cochlea

Layersof eye -

- Outer layer - Sclera and Cornea
- Middle layer- Choroid, Ciliary body, Iris, Pupil, lens
- Innermost layer- Retina



- Reflex action and Reflex arc

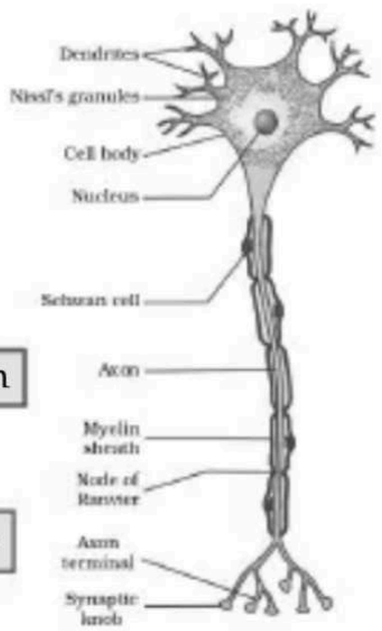


- Generation and conduction of nerve impulse
- Maintenance of resting potential
- Action potential
- Propagation of action potential

- Brain
- Fore brain
- Mid brain
- Hind brain
- Cerebrum
- Thalamus
- Hypothalamus
- Pons
- Cerebellum
- Medulla

- Human Neural System
- CNS
- Brain
- Spinal cord
- PNS
- Somatic
- Autonomic
- Parasympathetic
- Sympathetic
- Cranial nerves (12 pairs)
- Spinal nerves (31 pairs
- Nerve Fibres
- Afferent Fibres
- Efferent Fibres

- Neuron (Structural and functional unit of neural system)]
- Cytoplasm + typical cell organelles + Nissl's granules
- Cell body chain
- Dendrite chain
- Structure
- Axon chain
- Myelinated axon
- Non-myelinated axon
- Types of Neuron
- Multipolar
- Bipolar
- Unipolar



CHAPTER : 22 CHEMICAL COORDINATION AND INTEGRATION

Chemical Coordination and Integration

Hormones (Chemical messengers of the body)

Secretes

Types

Glands

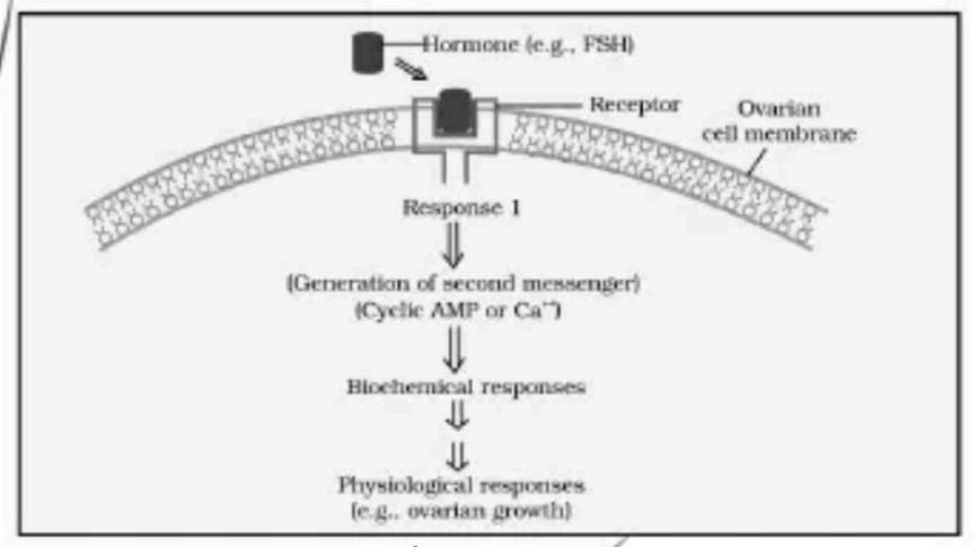
- Exocrine (With duct)
- Endocrine (Without duct)
- Heterocrine (Partly endocrine and partly exocrine)

Heart → ANF

Kidney → Renin, Erythropoietin and calcitriol

Gastrointestinal Tract→ Gastrin, GIP, Secretin, Cholecystokinin, Duocrinin, Villikinin, Enterocrinin

Hormones of Heart, Kidney and Gastrointestinal Tract



Releasing hormone

Inhibiting hormone

Hypothalamus

Adenohypophysis

Neurohypophysis

Pituitary

Melatonin

Pineal

T3

T4 (Thryoxine)

Thyroid

Thyrocalcitonin

Parathyroid

Parathyroid hormone

Thymosins

Thymus

Adrenal cortex

- Mineralocorticoids
- Glucocorticoids

Adrenal

Adrenal medulla

- Adrenaline
- Noradrenaline

Beta cells

Pancreas

Alpha cells

Insulin

Glucagon

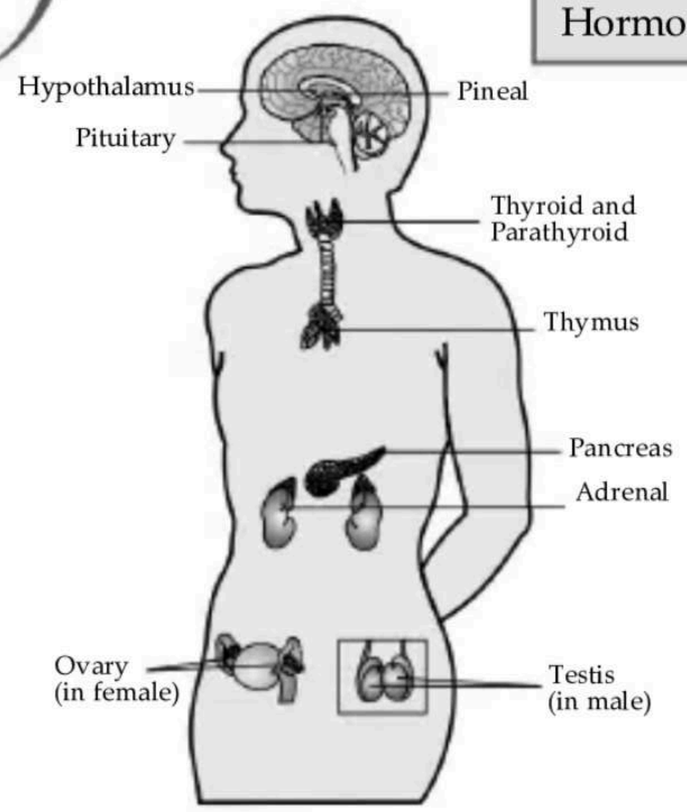
Testis

Testosterone

Ovary

- Estrogen
- Progesterone

Human Endocrine System



Mechanism of Hormone Action

Water soluble hormones

Lipid soluble hormones

